THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. In glass, the colour of light with the highest speed is 1

(a) violet **(b)** blue **(c)** red **(d)** green

2. Resistors of 2 ohm, 3 ohm and 6 ohm are connected in parallel. The equivalent resistance is 1

(a) 1 ohm **(b)** 2 ohm **(c)** 3 ohm **(d)** 11 ohm

3. A conductor of length L carrying current I is placed parallel to the field. The force is 1

(a) half of maximum **(b)** maximum

(c) equal to the weight **(d)** zero

4. Which one is a harmful effect of CFCs? 1

(a) Cooling the sun **(b)** Making plants grow

(c) Destroying the ozone layer **(d)** Increasing oxygen

5. Chips packets are filled with nitrogen to: 1

(a) prevent oxidation of the oil in the chips

(b) keep them warm

(c) make them react with air

(d) increase the weight

6. Which of the following is a strong base? 1

(a) NaOH **(b)** $Cu(OH)_2$ **(c)** $Al(OH)_3$ **(d)** $Mg(OH)_2$

7. The process of making oxide layer on aluminium thicker is: 1

(a) galvanisation **(b)** roasting **(c)** smelting **(d)** anodising

8. Carbon cannot form C_4^- because: 1

(a) it is heavy

(b) 6 protons cannot hold 10 electrons easily

(c) it is metal

(d) it has 8 electrons

9. Which of the following transports water and minerals upward? 1

(a) cortex **(b)** epidermis **(c)** xylem **(d)** phloem

10. A person has a swelling in the neck and is lethargic. The cause may be a deficiency of: 1

(a) iron **(b)** sodium **(c)** iodine **(d)** calcium

11. The structure that connects the embryo to the placenta is the 1

(a) urethra **(b)** cervix **(c)** oviduct **(d)** umbilical cord

12. Plants of a clone are grown in different soils and show different heights. This difference is 1

(a) inherited **(b)** a mutation of germ cells
(c) dominant **(d)** an acquired trait due to environment

13. A convex lens of f = 15 cm forms a real image at 60 cm. The object distance is 1

(a) 15 cm **(b)** 45 cm **(c)** 20 cm **(d)** 30 cm

14. Fine particles in the atmosphere scatter mainly 1

(a) red light **(b)** yellow light **(c)** no light **(d)** blue light

15. A 2 kW heater runs for 30 minutes. Energy consumed is 1

(a) 1 kWh **(b)** 2 kWh **(c)** 4 kWh **(d)** 60 kWh

16. A coil with 100 turns instead of 50 turns, other things same, gives 1

(a) a smaller induced emf **(b)** a larger induced emf
(c) no emf **(d)** the same induced emf

17. Composting is the process of 1

(a) burning plastic **(b)** recycling glass
(c) melting metals **(d)** converting vegetable waste into manure

18. The sum of coefficients in the balanced equation $\mathrm{Al} + \mathrm{O_2} \longrightarrow \mathrm{Al_2O_3}$ is: 1

(a) 6 **(b)** 7 **(c)** 9 **(d)** 5

19. The salt $\mathrm{CH_3COONa}$ in water is: 1

(a) amphoteric **(b)** acidic **(c)** basic **(d)** neutral

20. Which of the following is the correct order of reactivity? 1

(a) Mg > Zn > Fe > Cu **(b)** Fe > Cu > Mg > Zn
(c) Cu > Fe > Zn > Mg **(d)** Zn > Mg > Cu > Fe

21. The number of shared electron pairs in ethyne ($\mathrm{C_2H_2}$) is: 1

(a) 6 **(b)** 3 **(c)** 5 **(d)** 4

22. A leaf is kept in the dark for 24 hours and then tested with iodine. It will show: 1

(a) no blue-black colour **(b)** green colour
(c) red colour **(d)** blue-black colour

23. The impulse travels in a neuron from: 1

(a) axon to dendrite **(b)** cell body to dendrite
(c) synapse to axon **(d)** dendrite to axon end

24. A potato tuber has several eyes. Each eye can grow into a plant by 1

(a) vegetative propagation **(b)** regeneration
(c) budding **(d)** spore formation

25. Which of the following genotypes will produce only one type of gamete? 1

(a) RrYy **(b)** Rr **(c)** TT **(d)** Tt

26. A convex mirror has f = 20 cm. An object is at 30 cm. The image distance is 1

(a) 60 cm behind the mirror **(b)** 12 cm behind the mirror
(c) 20 cm behind **(d)** 12 cm in front

27. A myopic person's eyeball is too long. His eye forms the image of a distant object 1

(a) on the retina **(b)** in front of the retina
(c) behind the retina **(d)** at infinity

28. The V-I graph of a conductor has a slope of 5 ohm. If the current is 2 A, the voltage is 1

(a) 10 V **(b)** 2.5 V **(c)** 7 V **(d)** 0.4 V

29. When current in a solenoid is doubled (other factors same), the field inside 1

(a) becomes zero **(b)** remains the same **(c)** decreases **(d)** increases

30. If a top carnivore receives 10 J of energy through 3 transfers from producers, the energy at the producers is 1

(a) 1000 J **(b)** 10,000 J **(c)** 100,000 J **(d)** 100 J

31. Which of these is both a combination and an exothermic reaction? 1

(a) $\mathrm{CaO + H_2O \longrightarrow Ca(OH)_2}$ **(b)** $\mathrm{2\,AgCl \longrightarrow 2\,Ag + Cl_2}$ **(c)** $\mathrm{2\,H_2O \longrightarrow 2\,H_2 + O_2}$ **(d)** $\mathrm{CaCO_3 \longrightarrow CaO + CO_2}$

32. Hardening of plaster of Paris is due to: 1

(a) decomposition **(b)** formation of gypsum

(c) loss of water **(d)** oxidation

33. The formula of the compound formed between Mg and Cl is: 1

(a) MgCl **(b)** $\mathrm{Mg_2Cl_2}$ **(c)** $\mathrm{MgCl_2}$ **(d)** $\mathrm{Mg_2Cl}$

34. The reaction $\mathrm{C_2H_4 + H_2 \longrightarrow C_2H_6}$ with nickel is: 1

(a) addition reaction **(b)** esterification

(c) substitution reaction **(d)** combustion

35. A person who has run fast feels muscle pain due to: 1

(a) lactic acid accumulation **(b)** ethanol accumulation

(c) glucose excess **(d)** $\mathrm{CO_2}$ shortage

36. The hormone that is present in greater concentration in fruits and seeds is: 1

(a) ABA **(b)** cytokinin **(c)** gibberellin **(d)** auxin

37. Pollen grains of a flower reaching the stigma of another flower on the same plant is 1

(a) cross-pollination **(b)** fertilisation

(c) self-pollination **(d)** vegetative propagation

38. If a human sperm has 22 autosomes and the Y chromosome, the number of chromosomes is 1

(a) 44 **(b)** 23 **(c)** 22 **(d)** 46

39. The refractive index of glass is 1.5 and of water is 1.33. Light travels fastest in 1

(a) water **(b)** glass **(c)** both equally **(d)** neither

40. When a second inverted prism is placed in the path of dispersed light, we get 1

(a) only red **(b)** no light **(c)** white light **(d)** more colours

41. Two resistors of 4 ohm and 12 ohm in series are connected to a 16 V battery. The current is 1

(a) 16 A **(b)** 0.25 A **(c)** 4 A **(d)** 1 A

42. Which of the following reverses the force on a current-carrying conductor in a field? 1

(a) Reversing the current **(b)** Reducing the length

(c) Increasing the current **(d)** Heating it

43. The ozone layer absorbs 1

(a) infrared rays **(b)** visible light

(c) X-rays only **(d)** UV rays of the sun

44. Galvanised iron does not rust because zinc: 1

(a) is less reactive than iron
(b) is lighter than iron
(c) combines with iron
(d) is more reactive than iron and forms a protective layer

45. Which acid is used in car batteries? 1

(a) nitric acid **(b)** sulphuric acid
(c) acetic acid **(d)** hydrochloric acid

46. Which is NOT a method of preventing corrosion? 1

(a) galvanising **(b)** painting **(c)** alloying **(d)** heating in air

47. Diamond does not conduct electricity because: 1

(a) no free electrons **(b)** it is metal
(c) it is soft **(d)** it is liquid

48. Loss of water from leaves creates: 1

(a) blood pressure **(b)** transpiration pull
(c) osmosis only **(d)** root pressure only

49. Fight or flight response: Heart rate increases because of: 1

(a) insulin **(b)** oestrogen **(c)** thyroxine **(d)** adrenaline

50. Where do the sperms mix with the secretions of the glands? 1

(a) In the urethra / vas deferens region **(b)** In the uterus lining
(c) In the ovary **(d)** In the placenta

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND